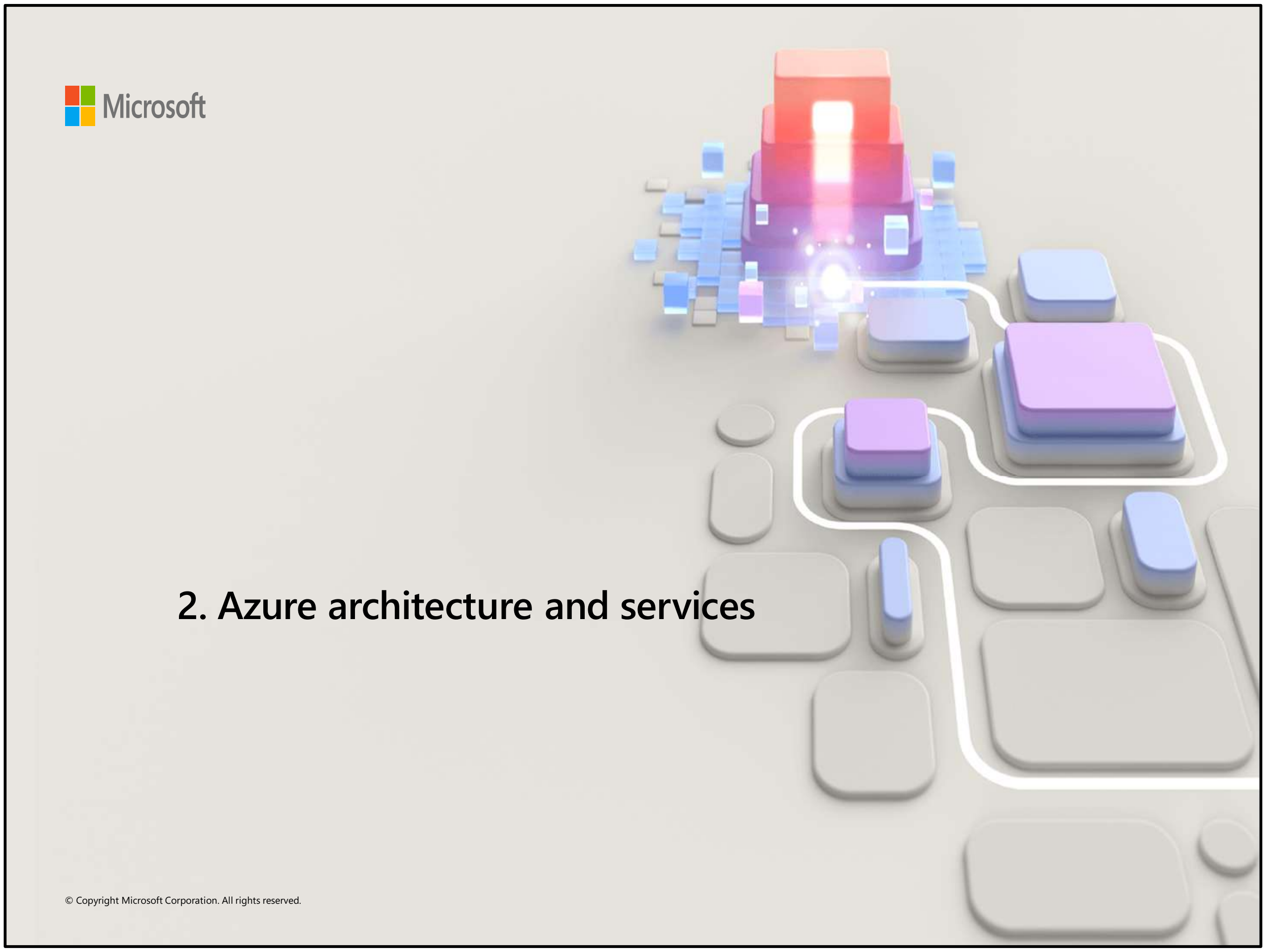


Learn learning path:
https://learn.microsoft.com/training/paths/azure-fundamentals-describe-azure-architecture-services/

Learning path 02—outline

You will learn the following concepts:

1. **Azure architectural components**
   - Regions and availability zones
   - Subscriptions and resource groups
2. **Compute and networking**
   - Compute types
   - Application hosting
   - Virtual networking
3. **Storage**
   - Storage services
   - Redundancy options
   - File management and migration
4. **Identity, access, and security**
   - Directory services
   - Authentication methods
   - Security models

Azure accounts

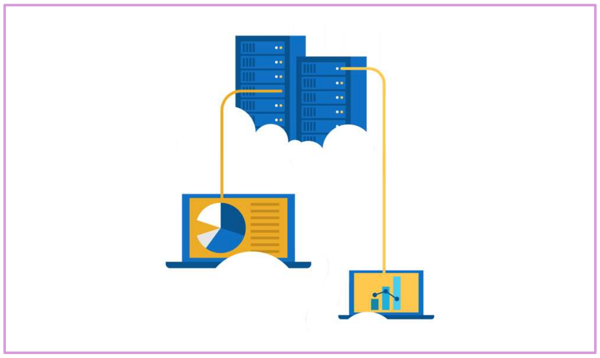
- Azure account
- Azure free account
- Azure free student account
- Microsoft Learn sandbox

https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/3-get-started-azure-accounts

Walkthrough—create an Azure account

Create a free Azure account

1. Create a free Azure free account.



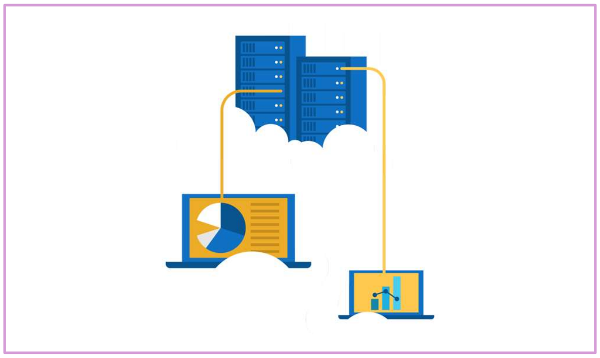
https://learn.microsoft.com/training/modules/create-an-azure-account/

Most exercises will be in the Learn Sandbox—in the last module, there's a portion that needs an Azure account or can be converted to Demo. This is an optional exercise to help the students create free accounts. Note: They will need a credit card for this exercise.

Exercise—explore the Learn sandbox

Explore the Learn sandbox

1. Activate the sandbox
2. Use PowerShell
3. Shift to BASH
4. Shift to Azure Interactive mode
5. Navigate the portal



https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/4-exercise-explore-learn-sandbox

NOTE: The Learn sandbox doesn't require an Azure account. It will work with a Microsoft account, if you skipped the previous walkthrough.

Azure architectural components

https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/1-introduction

Core Azure architectural components—objective domain

- Describe Azure regions, region pairs, and sovereign regions.
- Describe Availability Zones.
- Describe Azure datacenters.
- Describe Azure resources and Resource Groups.
- Describe subscriptions.
- Describe management groups.
- Describe the hierarchy of resource groups, subscriptions, and management groups.

https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/1-introduction

Regions

Azure offers more global regions than any other cloud provider with 60-plus regions representing over 140 countries

- Regions are made up of one or more datacenters in close proximity.
- They provide flexibility and scale to reduce customer latency.
- Regions preserve data residency with a comprehensive compliance offering.

https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/5-describe-azure-physical-infrastructure

A region represents a collection of datacenters.

They provide flexibility and scale.

Regions preserve data residency.

You can select regions close to your users.

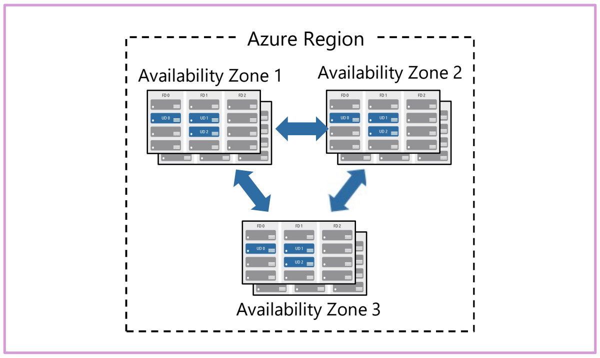
Be aware of region deployment availability.

There are global services that are region independent.

A list of regions and their locations is available at https://azure.microsoft.com/explore/global-infrastructure/data-residency/#overview

Availability zones

- Provide protection against downtime due to datacenter failure.
- Physically separate datacenters within the same region.
- Each datacenter is equipped with independent power, cooling, and networking.
- Connected through private fiber-optic networks.



https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/5-describe-azure-physical-infrastructure

Physically separate locations within an Azure region.
Takes availability sets to the next level
Includes one or more datacenters, equipped with independent power, cooling, and networking.
Acts as an isolation boundary.
If one availability zone goes down, the other continues working.

More details about Availability Zones in Azure are available at https://learn.microsoft.com/azure/reliability/availability-zones-overview

Region pairs

- At least 300 miles of separation between region pairs.
- Automatic replication for some services.
- Prioritized region recovery in the event of outage.
- Updates are rolled out sequentially to minimize downtime.
- Web link: https://aka.ms/PairedRegions

Region		Region
North Central US	⟷	South Central US
East US	⟷	West US
West US 2	⟷	West Central US
US East 2	⟷	Central US
Canada Central	⟷	Canada East
North Europe	⟷	West Europe
UK West	⟷	UK South
Germany Central	⟷	Germany Northeast
South East Asia	⟷	East Asia
East China	⟷	North China
Japan East	⟷	Japan West
Australia Southeast	⟷	Australia East
India South	⟷	India Central
Brazil South (Primary)	⟷	South Central US

https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/5-describe-azure-physical-infrastructure

Each Azure region is paired with another region.

Azure prefers at least 300 miles of separation between datacenters in a regional pair.

Some services provide automatic replication to the paired region.

In an outage, recovery of one region is prioritized out of every pair.

Azure system updates are rolled out to paired regions sequentially (not at the same time).

For a list of geographies, regions, region-pairs, and other details, go to https://azure.microsoft.com/explore/global-infrastructure/geographies/#overview

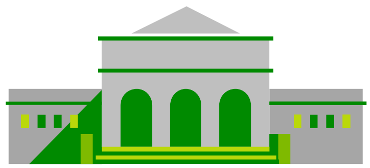
For a full list of region pairs, go to https://learn.microsoft.com/azure/reliability/cross-region-replication-azure#what-are-paired-regions

Azure sovereign regions (US government services)

Meets the security and compliance needs of US federal agencies, state and local governments, and their solution providers.

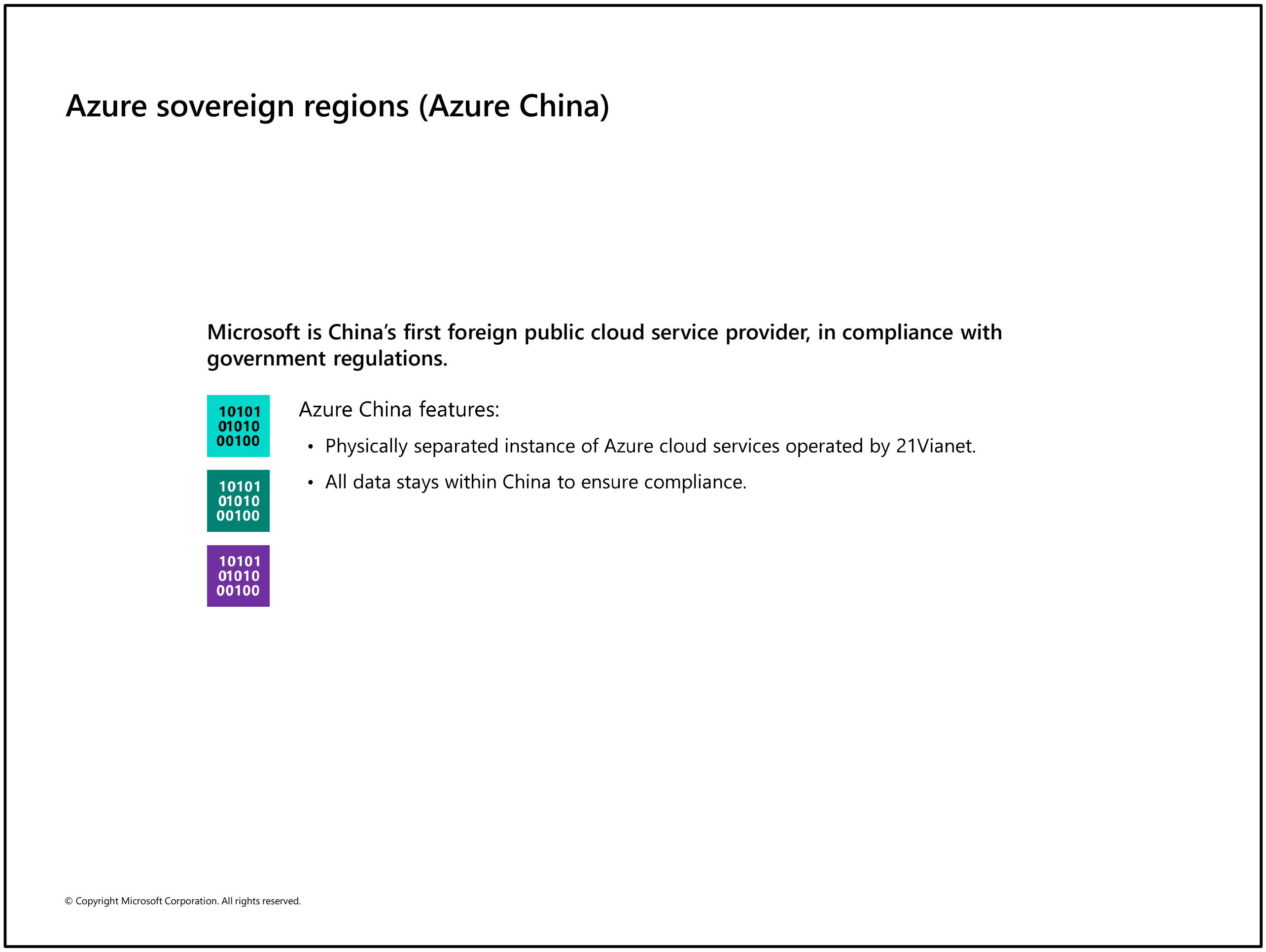
Azure government:

- Separate instance of Azure.
- Physically isolated from non-US government deployments.
- Accessible only to screened, authorized personnel.



https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/5-describe-azure-physical-infrastructure

Azure Government - https://azure.microsoft.com/en-us/global-infrastructure/government/



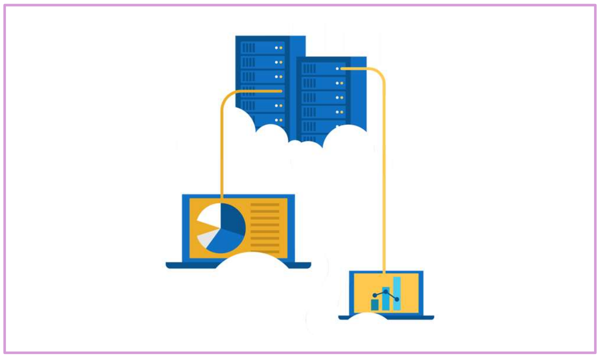
https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/5-describe-azure-physical-infrastructure

Azure China 21Vianet - https://learn.microsoft.com/azure/china/

Walkthrough—explore the Azure global infrastructure

Explore the Azure global infrastructure

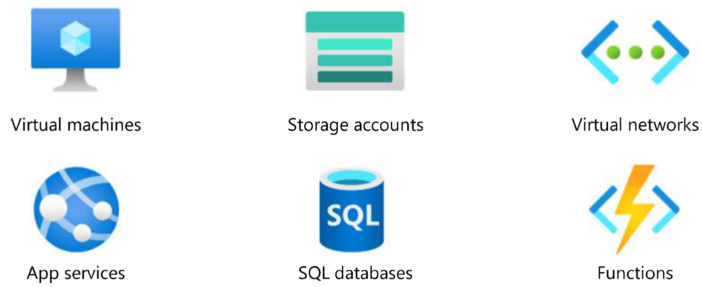
1. Select **Explore the Globe** (after intro).
2. Notice the different icons (geography, regions, points of presence (PoP), and so on).
3. Find your location on the globe, then find the nearest PoP and region to your location.



https://datacenters.microsoft.com/globe/

Azure resources

Azure **resources** are components like storage, virtual machines, and networks that are available to build cloud solutions.

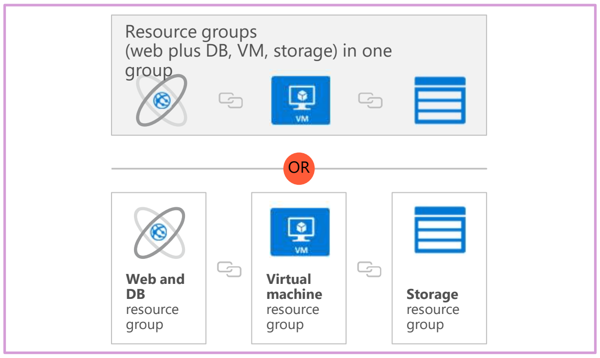


https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure

Resource groups

A **resource group** is a container you use to manage and aggregate resources in a single unit.

- Resources can exist in only one resource group.
- Resources can exist in different regions.
- Resources can be moved to different resource groups.
- Applications can utilize multiple resource groups.



https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure

Containers for multiple resources that share the same life cycle.

Aggregates resources into a single manageable unit.

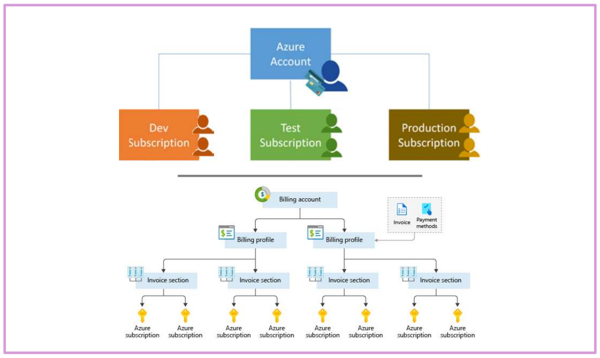
Every Azure resource must exist in one (and only one) resource group.

Secure at the resource group
(or resource) level—using role-based access control (RBAC).

Azure subscriptions

An Azure subscription provides you with authenticated and authorized access to Azure accounts.

- **Billing boundary:** Generate separate billing reports and invoices for each subscription.
- **Access control boundary:** Manage and control access to the resources that users can provision with specific subscriptions.



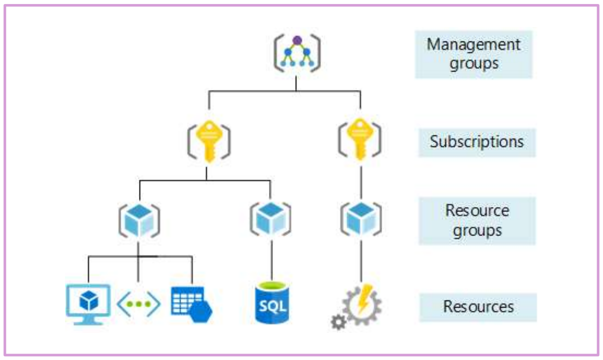
https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure

An account can have one or multiple subscriptions.

Azure subscription offers: https://azure.microsoft.com/en-us/support/legal/offer-details/

Management groups

- Management groups can include multiple Azure subscriptions.
- Subscriptions inherit conditions applied to the management group.
- 10,000 management groups can be supported in a single directory.
- A management group tree can support up to six levels of depth.



https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure

Management groups can include multiple Azure subscriptions.

Subscriptions inherit conditions applied to the management group.

10,000 management groups can be supported in a single directory.

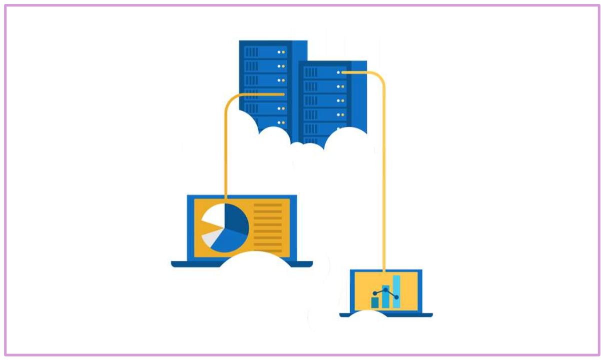
A management group tree can support up to six levels of depth.

Management groups: https://learn.microsoft.com/azure/governance/management-groups/

Exercise—create an Azure resource

Create an Azure resource and monitor the resource group for the required resources being created in the same group.

1. Create a virtual machine.
2. Monitor the resource group.



https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/7-exercise-create-azure-resource/

Lab to create an Azure resource. A key component of this lab is noticing that when a VM is created, Azure automatically creates the necessary components (network, storage, and so on) and groups them automatically in the same resource group.

Compute and networking

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/

Compute and networking—objective domain

Describe the benefits and usage:

- Compare compute types, including container instances, virtual machines, and functions.
- Describe virtual machine options, including virtual machines (VMs), virtual machine scale sets, virtual machine availability sets, and Azure Virtual Desktop.
- Describe the resources required for virtual machines.
- Describe application hosting options, including Azure Web Apps, containers, and virtual machines.
- Describe virtual networking, including the purpose of Azure Virtual Networks, Azure virtual subnets, peering, Azure DNS, VPN Gateway, and ExpressRoute.
- Define public and private endpoints.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/1-introduction

Azure compute services

Azure **compute** is an on-demand service that provides computing resources such as disks, processors, memory, networking, and operating systems.

Virtual Machines

App Services

Container Instances

Azure Kubernetes Services (AKS)

Azure Virtual Desktop

https://azure.microsoft.com/products/category/compute/

Azure virtual machines

Azure **virtual machines (VMs)** are software emulations of physical computers.

- Includes virtual processor, memory, storage, and networking.
- IaaS offering that provides total control and customization.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/2-virtual-machines

Development and test: Azure VMs offer a quick and easy way to create a computer with specific configurations required to code and test an application.

Applications in the cloud: Because demand for your application can fluctuate, it might make economic sense to run it on a VM in Azure. You pay for extra VMs when you need them and shut them down when you don't.

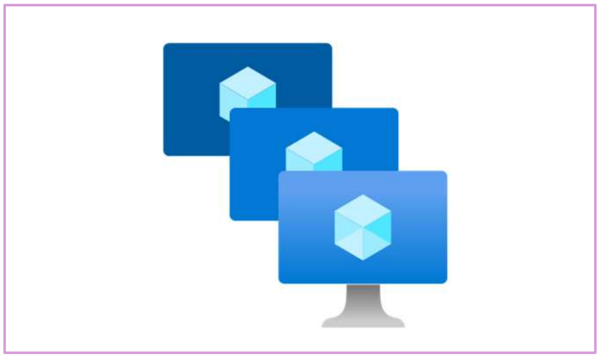
Extended datacenter: Virtual machines in an Azure virtual network can easily be connected to your organization's network.

Azure virtual machines: https://azure.microsoft.com/en-us/services/virtual-machines/

VM scale sets

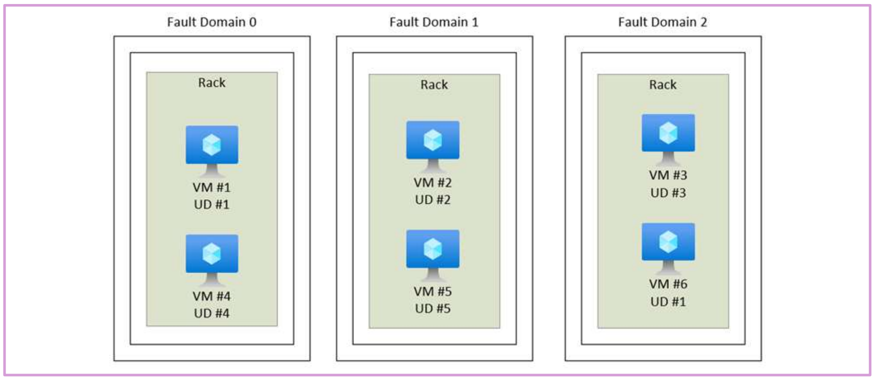
Scale sets provide a load-balanced opportunity to automatically scale resources.

- Scale out when resource needs increase.
- Scale in when resource needs are lower.



https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/2-virtual-machines

VM availability sets



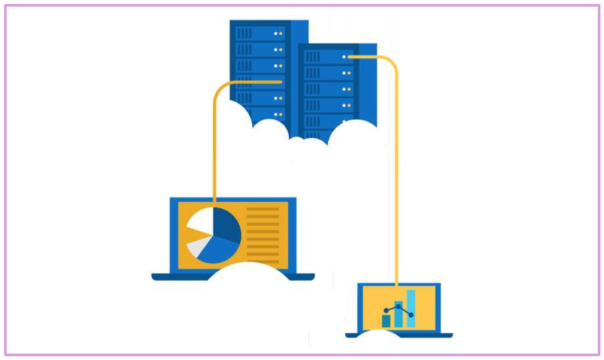
https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/2-virtual-machines

Update domain keeps VMs grouped if they can be rebooted at the same time without causing an outage.
Fault domains group VMs based on common power and networking.

Exercise—create a virtual machine

Create a virtual machine in the Azure portal, connect to the virtual machine, install the web server role, and test.

1. Create the virtual machine.
2. Install the web server package.



https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/3-exercise-create-azure-virtual-machine

Azure Virtual Desktop

Azure Virtual Desktop is a desktop and app virtualization that runs in the cloud.

- Create a full desktop virtualization environment without having to run additional gateway servers.
- Reduce risk of resource being left behind.
- True multisession deployments.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/4-virtual-desktop

https://learn.microsoft.com/azure/virtual-desktop/overview

Azure container services

Azure **containers** provide a lightweight, virtualized environment that does not require operating system management, and can respond to changes on demand.

Azure Container Instances: A PaaS offering that runs a container or pod of containers in Azure.

Azure Container Apps: A PaaS offering, like container instances, that can load balance and scale.

Azure Kubernetes Service: An orchestration service for containers with distributed architectures and large volumes of containers.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/5-containers

Containers are a virtualization environment. However, unlike virtual machines, you do not manage an operating system. Containers are meant to be lightweight, and are designed to be created, scaled out, and stopped dynamically.

Azure Container Instances: https://azure.microsoft.com/en-us/services/container-instances/

Azure Kubernetes Service: https://azure.microsoft.com/en-us/services/kubernetes-service/

Azure Functions

Azure Functions: A PaaS offering that supports serverless compute operations. Event-based code runs when called without requiring server infrastructure during inactive periods.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/6-functions

Serverless computing is the evolution of cloud platforms in the direction of pure cloud native code. Serverless brings developers closer to business logic while insulating them from infrastructure concerns. It's a pattern that doesn't imply "no server" but rather, "less server." Serverless code is event-driven. Code may be triggered by anything from a traditional HTTP web request to a timer or the result of uploading a file. The infrastructure behind serverless allows for instant scale to meet elastic demands and offers microbilling to truly "pay for what you use." Serverless requires a new way of thinking and approach to building applications and isn't the right solution for every problem.

Azure Functions is code that runs your service and not the underlying platform or infrastructure. Creates infrastructure based on an event.

Comparing Azure compute options

Virtual machines

- Cloud-based server that supports either Windows or Linux environments.
- Useful for lift-and-shift migrations to the cloud.
- Complete operating system package, including the host operating system.

Virtual Desktop

- Provides a cloud-based personal computer Windows desktop experience.
- Dedicated applications to connect and use, or accessible from any modern browser.
- Multiclient login allows multiple users to log into the same machine at the same time.

Containers

- Lightweight, miniature environment well suited for running microservices.
- Designed for scalability and resiliency through orchestration.
- Applications and services are packaged in a container that sits on top of the host operating system. Multiple containers can sit on one host OS.

Azure App Services

Azure **App Services** is a fully managed platform to build, deploy, and scale web apps and APIs quickly.

- Works with .NET, .NET Core, Node.js, Java, Python, or php.
- PaaS offering with enterprise-grade performance, security, and compliance requirements.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/7-describe-application-hosting-options

Azure networking services

Azure Virtual Network (VNet) enables Azure resources to communicate with each other, the internet, and on-premises networks.

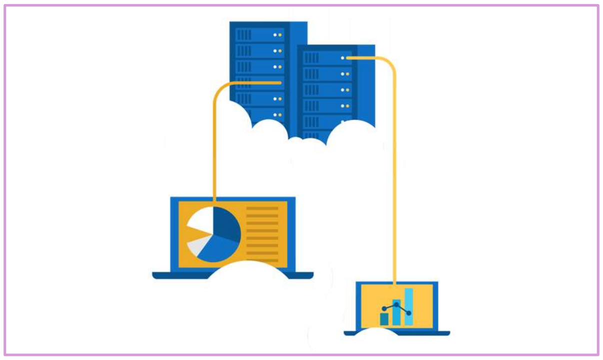
- Public endpoints, accessible from anywhere on the internet.
- Private endpoints, accessible only from within your network.
- Virtual subnets segment your network to suit your needs.
- Network peering connects your private networks directly together.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/8-virtual-networking

Walkthrough—configure network access

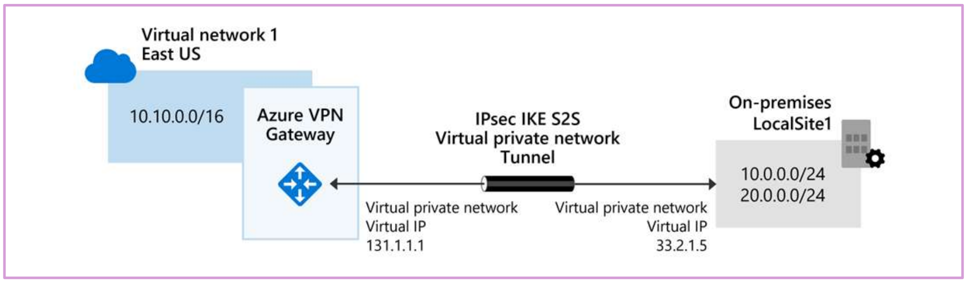
Configure public access to the virtual machine created earlier.

1. Verify currently open ports.
2. Create a network security group
3. Configure HTTP access (port 80)
4. Test the connection.



https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/9-exercise-configure-network-access/

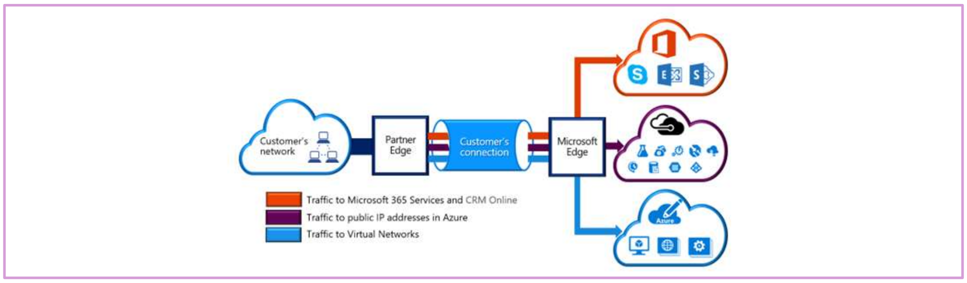
Azure networking services: VPN Gateway



VPN Gateway is used to send encrypted traffic between an Azure virtual network and an on-premises location over the public internet.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/10-virtual-private-networks

Azure networking services: ExpressRoute



ExpressRoute extends on-premises networks into Azure over a private connection that is facilitated by a connectivity provider.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/11-expressroute/

Azure DNS

- Reliability and performance by leveraging a global network of DNS name servers using Anycast networking.
- Azure DNS security is based on Azure resource manager, enabling role-based access control and monitoring and logging.
- Ease of use for managing your Azure and external resources with a single DNS service.
- Customizable virtual networks allow you to use private, fully customized domain names in your private virtual networks.
- Alias records support alias record sets to point directly to an Azure resource.

https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/12-domain-name-system

Storage

https://learn.microsoft.com/training/modules/describe-azure-storage-services/

Storage—objective domain

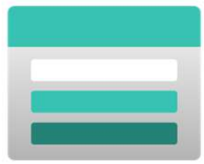
Describe the benefits and usage

- Compare Azure storage services.
- Describe storage tiers.
- Describe redundancy options.
- Describe storage account options and storage types.
- Identify options for moving files, including AzCopy, Azure Storage Explorer, and Azure File Sync.
- Describe migration options, including Azure Migrate and Azure Data Box.

https://learn.microsoft.com/training/modules/describe-azure-storage-services/1-introduction

Storage accounts

- Must have a globally unique name.
- Provide over-the-internet access worldwide.
- Determine storage services and redundancy options.



https://learn.microsoft.com/training/modules/describe-azure-storage-services/2-accounts

Storage redundancy

Redundancy configuration	Deployment	Durability
Locally redundant storage (LRS)	Single datacenter in the primary region	11 nines
Zone-redundant storage (ZRS)	Three availability zones in the primary region	12 nines
Geo-redundant storage (GRS)	Single datacenter in the primary and secondary region	16 nines
Geo-zone-redundant-storage (GZRS)	Three availability zones in the primary region and a single datacenter in the secondary region	16 nines

https://learn.microsoft.com/training/modules/describe-azure-storage-services/3-redundancy

Azure storage services

Azure Blob: Optimized for storing massive amounts of unstructured data, such as text or binary data.

Azure Disk: Provides disks for virtual machines, applications, and other services to access and use.

Azure Queue: Message storage service that provides storage and retrieval for large amounts of messages, each up to 64 KB.

Azure Files: Sets up a highly available network file share that can be accessed by using the Server Message Block protocol.

Azure Tables: Provides a key/attribute option for structured nonrelational data storage with a schema-less design.

https://learn.microsoft.com/training/modules/describe-azure-storage-services/4-describe-azure-storage-services

Storage service public endpoints

Storage service	Public endpoint
Blob Storage	https://<storage-account-name>.blob.core.windows.net
Data Lake Storage Gen2	https://<storage-account-name>.dfs.core.windows.net
Azure Files	https://<storage-account-name>.file.core.windows.net
Queue Storage	https://<storage-account-name>.queue.core.windows.net
Table Storage	https://<storage-account-name>.table.core.windows.net

Azure storage access tiers

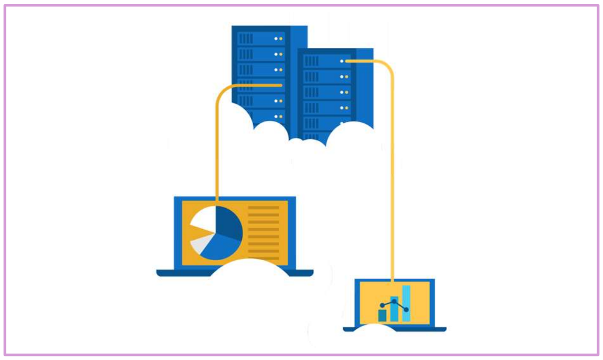
Hot	Cool	Cold	Archive
Optimized for storing data that is accessed frequently.	Optimized for storing data that is infrequently accessed and stored for at least 30 days.	Optimized for storing data that is infrequently accessed and stored for at least 90 days.	Optimized for storing data that is rarely accessed and stored for at least 180 days with flexible latency requirements.

https://learn.microsoft.com/training/modules/describe-azure-storage-services/4-describe-azure-storage-services

Exercise—create a storage blob

Create a storage account with a blob storage container. Work with blob files.

1. Create a storage account.
2. Create a blob container.
3. Upload and access a blob.



https://learn.microsoft.com/training/modules/describe-azure-storage-services/5-exercise-create-storage-blob

Azure Migrate

- Unified migration platform.
- Range of integrated and standalone tools.
- Assessment and migration.

https://learn.microsoft.com/training/modules/describe-azure-storage-services/6-identify-azure-data-migration-options

Azure Data Box

- Store up to 80 terabytes of data.
- Move your disaster recovery backups to Azure.
- Protect your data in a rugged case during transit.
- Migrate data out of Azure for compliance or regulatory needs.
- Migrate data to Azure from remote locations with limited or no connectivity.

https://learn.microsoft.com/training/modules/describe-azure-storage-services/6-identify-azure-data-migration-options

File management options

AzCopy

- Command-line utility.
- Copy blobs or files to or from your storage account.
- One-direction synchronization.

Azure Storage Explorer

- Graphical user interface (similar to Windows Explorer).
- Compatible with Windows, MacOS, and Linux.
- Uses AzCopy to handle file operations.

Azure File Sync

- Synchronizes Azure and on-premises files in a bidirectional manner.
- Cloud tiering keeps frequently accessed files local, while freeing up space.
- Rapid reprovisioning of failed local server (install and resync).

https://learn.microsoft.com/training/modules/describe-azure-storage-services/7-identify-azure-file-movement-options

Identity, access, and security

https://learn.microsoft.com/training/modules/describe-azure-identity-access-security

Identity, access, and security—objective domain

Describe the benefits and usage

- Describe directory services in Azure, including Microsoft Entra ID and Microsoft Entra Domain Services.
- Describe authentication methods in Azure, including single sign-on (SSO), multifactor authentication (MFA), and passwordless.
- Describe external identities and guest access in Azure.
- Describe Entra Conditional Access.
- Describe role-based access control (RBAC).
- Describe the concept of Zero Trust.
- Describe the purpose of the defense in depth model.
- Describe the purpose of Microsoft Defender for Cloud.

https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/1-introduction

Microsoft Entra ID

Microsoft Entra ID is Microsoft Azure's cloud-based identity and access management service.

- Authentication (employees sign in to access resources).
- Single sign-on (SSO).
- Application management.
- Business to Business (B2B).
- Device management.

https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/2-directory-services

Azure AD - https://azure.microsoft.com/en-us/services/active-directory/

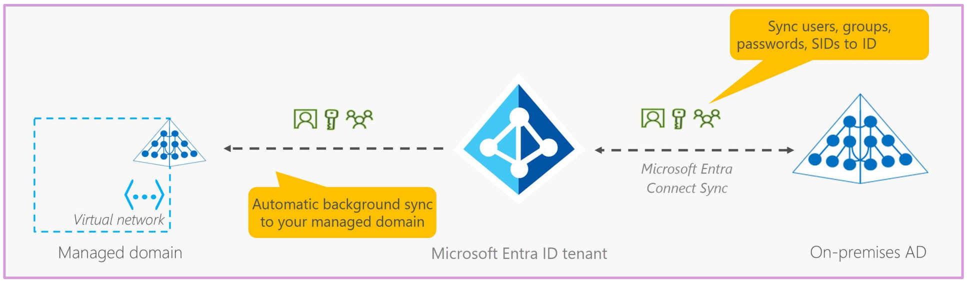
Azure tenant

A dedicated and trusted instance of Azure AD that's automatically created when your organization signs up for a Microsoft cloud service subscription, such as Microsoft Azure, Microsoft Intune, or Office 365. An Azure tenant represents a single organization.

Azure AD directory

Each Azure tenant has a dedicated and trusted Azure AD directory. The Azure AD directory includes the tenant's users, groups, and apps and is used to perform identity and access management functions for tenant resources.

Microsoft Entra Domain Services



- Gain the benefit of cloud-based domain services without managing domain controllers.
- Run legacy applications (that can't use modern auth standards) in the cloud.
- Automatically sync from Microsoft Entra ID.

https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/2-directory-services

Microsoft Entra ID: https://www.microsoft.com/security/business/identity-access/microsoft-entra-id

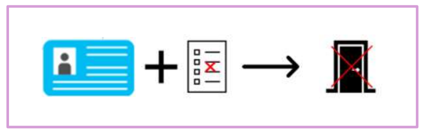
Compare authentication and authorization

Authentication

- Identifies the person or service seeking access to a resource.
- Requests legitimate access credentials.
- Basis for creating secure identity and access control principles.

Authorization

- Determines an authenticated person's or service's level of access.
- Defines which data they can access, and what they can do with it.



https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/3-authentication-methods

Authentication is sometimes shortened to *AuthN*, and authorization is sometimes shortened to *AuthZ*.

Authentication
- Identifies the person or service seeking access to a resource.
- Requests legitimate access credentials.
- Basis for creating secure identity and access control principles.

Authorization
- Determines an authenticated person's or service's level of access.
- Defines which data they can access, and what they can do with it.

Multifactor authentication

Provides additional security for your identities by requiring two or more elements for full authentication.

- Something you know ← → Something you possess ← → Something you are

https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/3-authentication-methods

Discuss what could qualify for each of the items listed:

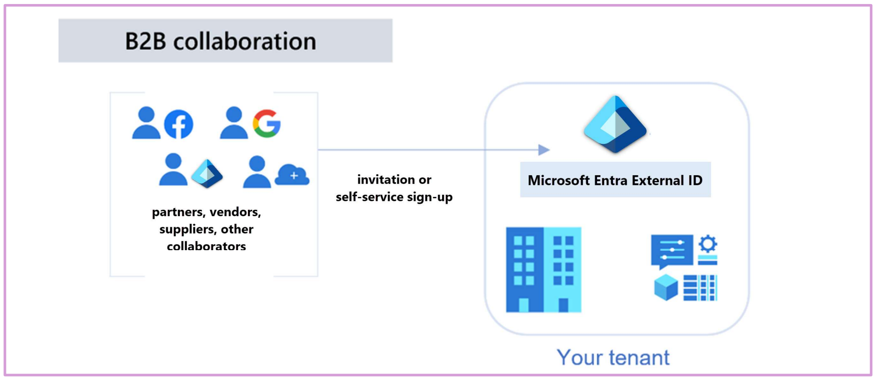
Something you know: This could be a password or the answer to a security question.

Something you possess: This might be a mobile app that receives a notification, or a token-generating device.

Something you are: This is typically some sort of biometric property, such as a fingerprint or face scan used on many mobile devices.

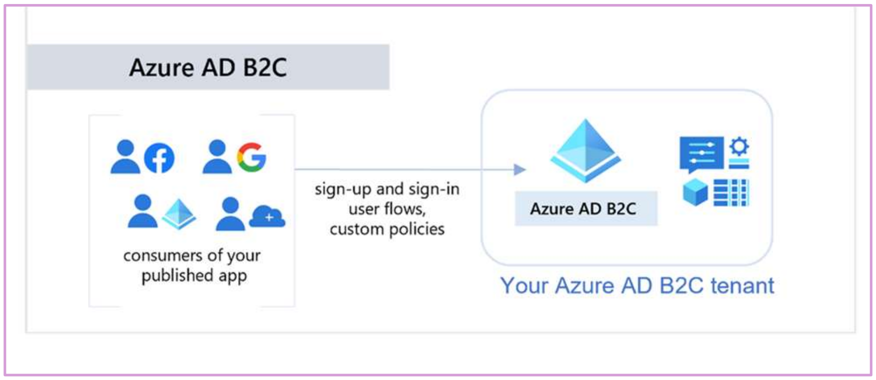
MFA: https://learn.microsoft.com/azure/active-directory/authentication/concept-mfa-howitworks

Microsoft Entra External ID B2B



https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/4-external-identities

Azure AD External Identities B2C

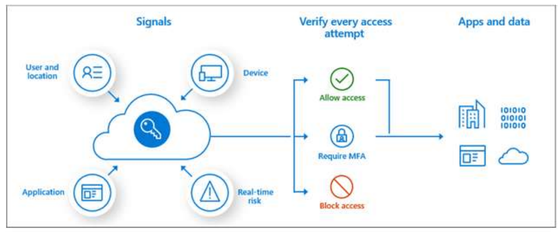


https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/4-external-identities

Conditional Access

Conditional Access is used to bring signals together, to make decisions, and enforce organizational policies.

- User or group membership
- IP location
- Device
- Application
- Risk detection



https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/5-conditional-access

https://learn.microsoft.com/azure/active-directory/conditional-access/overview

The modern security perimeter now extends beyond an organization's network to include user and device identity. Organizations can utilize these identity signals as part of their access control decisions.

Conditional Access is the tool used to bring signals together, to make decisions, and enforce organizational policies. Conditional Access is at the heart of the new identity driven control plane.

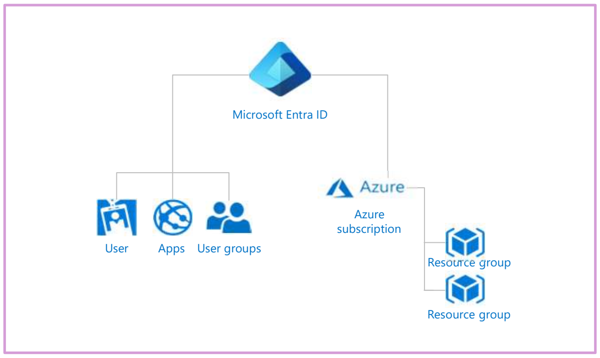
At their simplest, Conditional Access policies are if-then statements. If a user wants to access a resource, then they must complete an action. Example: A payroll manager wants to access the payroll application and is required to perform multifactor authentication.

Administrators are faced with two primary goals:

- Empower users to be productive wherever and whenever.
- Protect the organization's assets.

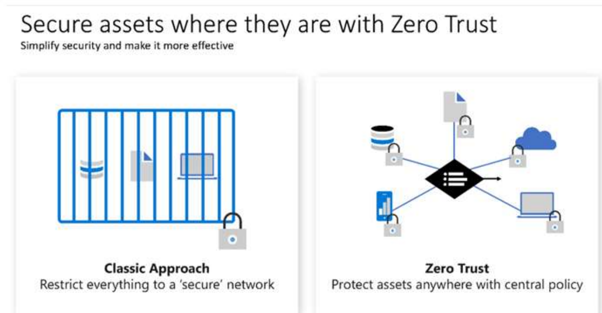
Role-based access control

- Fine-grained access management.
- Segregate duties within the team and grant only the amount of access to users that they need to perform their jobs.
- Enables access to the Azure portal and controlling access to resources.



https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/6-role-based-access-control
Azure RBAC: https://learn.microsoft.com/azure/role-based-access-control/overview

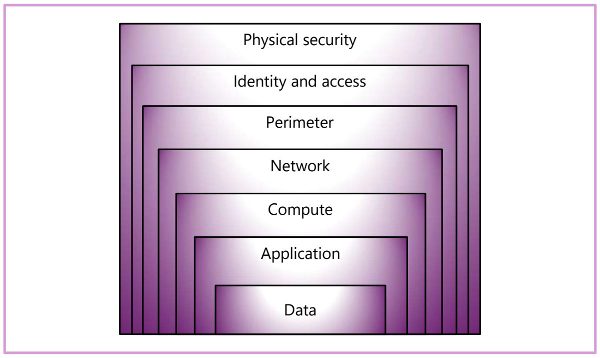
Zero Trust



https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/7-describe-zero-trust-model

Defense in depth

- A layered approach to securing computer systems.
- Provides multiple levels of protection.
- Attacks against one layer are isolated from subsequent layers.

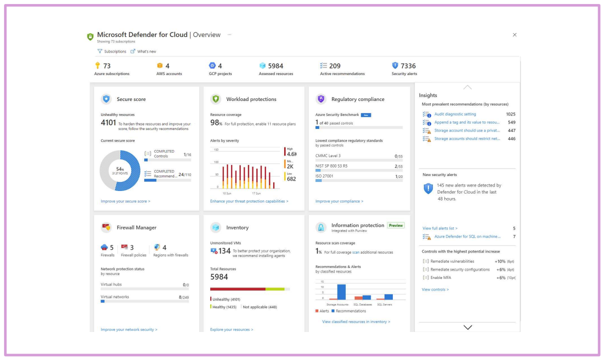


https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/8-describe-defense-depth

Microsoft Defender for Cloud

Microsoft Defender for Cloud is a monitoring service that provides threat protection across both Azure and on-premises datacenters.

- Provides security recommendations.
- Detect and block malware.
- Analyze and identify potential attacks.
- Just-in-time access control for ports.



https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/9-describe-microsoft-defender-for-cloud

Microsoft Defender for Cloud is a unified infrastructure security management system that strengthens the security posture of your datacenters and provides advanced threat protection across your hybrid workloads in the cloud—whether they're in Azure or not—as well as on-premises.

- Strengthen security posture: Defender for Cloud assesses your environment and enables you to understand the status of your resources, and whether they are secure.
- Protect against threats: Defender for Cloud assesses your workloads and raises threat prevention recommendations and security alerts.
- Get secure faster: With Defender for Cloud, everything is done in cloud speed. Because it's natively integrated, deployment is easy, giving you auto-provisioning and protection with Azure services.
- Policy compliance: Defender for Cloud is built on top of Azure Policy controls so you can set and monitor your policies to run on management groups, across subscriptions, and even for a whole tenant.
- Security alerts: Defender for Cloud automatically collects, analyzes, and integrates log data from your Azure resources, like firewall and endpoint protection, to detect real threats. The list of prioritized security alerts is shown in Microsoft Defender for Cloud along with the information you need to quickly investigate and remediate an attack.
- Secure score: Defender for Cloud continually assesses your resources for security issues, then aggregates all the findings into a single score so you can tell your current security situation.

Microsoft Defender for Cloud: https://azure.microsoft.com/services/defender-for-cloud/

Learning path 02 review

Microsoft Learn Modules (learn.microsoft.com/training)

- Physical and management infrastructure of Microsoft Azure
- Compute and networking services
- Storage services
- Identity, access, and security

https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/9-summary
https://learn.microsoft.com/training/modules/describe-azure-compute-networking-services/14-summary
https://learn.microsoft.com/training/modules/describe-azure-storage-services/9-summary
https://learn.microsoft.com/training/modules/describe-azure-identity-access-security/11-summary